

IMPERIAL COLLEGE BUSINESS SCHOOL

MSc Risk Management and Financial Engineering
Applied Project Report

RQMC-Enhanced Brownian-Bridge Conditioning

Pricing and Greeks of a Three-Asset
Worst-of Autocallable

FINANCIAL ENGINEERING · APPLIED QUANTITATIVE ANALYSIS

Advanced Monte Carlo Methods for Pricing and Sensitivity Analysis of Path-Dependent
Options

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Project AI Usage and Reflections (Please insert this after the cover page)

Students should explain how AI was used to support the development of their report and how its outputs were interpreted. They should include this template in their final report. As a guide, they should aim for 250-300 words per section.

Section 1: How I Used AI

Describe the tools used in your report (e.g., ChatGPT, Copilot, other platforms).

Explain the purpose (e.g., drafting, summarising, generating ideas, coding, interpreting data, proofreading).

I used OpenAI ChatGPT, including its Codex workspace, primarily as a coding assistant. I first identified the research question, located the market data and academic literature, read the relevant papers, and extracted the ideas that were applicable to my project. Based on this work, I specified the contracts, numerical methods, validation benchmarks, parameter regimes, and questions that the computational experiments needed to answer.

I then used AI to help translate these specifications into Python and Jupyter implementations. This included code for Monte Carlo and scrambled-Sobol simulation, correlated geometric Brownian motion, Brownian-bridge survival probabilities, finite-difference Greeks with common random numbers, fair-coupon root solving, repeated experiments, diagnostic tables, figure generation, and CPU/GPU comparison. AI was also useful for debugging, explaining unfamiliar parts of proposed code, and suggesting checks for numerical edge cases. However, I did not assume that generated code was correct. I executed it, inspected intermediate results, compared it with analytical or published benchmarks, and modified or rejected it when it did not meet the project's validation criteria.

The literature review, data collection, interpretation, and report writing remained my work. I decided which arguments from each paper were relevant and how they connected to the project's central question. Similarly, the code could generate many tables and figures, but I selected only the evidence that contributed directly to the report's main line of argument. For example, I prioritised results showing numerical validity, Greek stability, and time-to-accuracy trade-offs, while excluding technically correct but repetitive diagnostics. I wrote the report text myself, using AI only occasionally to discuss coding issues, presentation options, or clarity.

Section 2: AI-Assisted vs. Non AI-Assisted Work

Compare AI-generated content with the work you would have otherwise produced without AI and your own interpretation of outputs.

Describe critically at least two instances where you verified, corrected, or rejected AI output. For each, explain how you checked it (e.g., source, calculation, test), what you changed, and whether the AI use ultimately improved your work.

Without AI, I would have followed the same research design and used the same data, papers, and mathematical methods, but implementing and debugging the experiments would have taken substantially longer. AI mainly accelerated the conversion of my research specifications into modular code and made it easier to generate robustness checks. The non-AI contribution was to determine what should be tested, judge whether an output was credible, interpret its relevance, and construct the final written argument.

One verification concerned the simulation engines. AI-assisted code for Monte Carlo, scrambled Sobol, finite-difference Greeks, and Brownian-bridge conditioning was not used directly as evidence. I first tested the RQMC engine on a European option with Black–Scholes values for Price, Delta, Vega, and Gamma. I then tested the barrier implementation against a single-asset Brownian-bridge benchmark and checked the direct autocallable engine through cash-flow decomposition, probability-mass identities, fair-coupon repricing, and published contract examples. These checks supported some outputs but also showed that improved sampling did not automatically stabilise Gamma. I therefore reported that limitation instead of adopting a stronger AI-suggested interpretation.

A second instance involved Lee et al.'s logistic approximation for the trivariate exit probability. The local reproduction result, 0.165739 against the paper's value of approximately 0.1631, initially appeared successful. I designed broader out-of-sample tests across seeds and parameter regimes and compared the approximation with a high-budget nested reference. Its RMSE was 0.003706, above my pre-specified tolerance of 0.002500. I therefore rejected the logistic surrogate as the production estimator and reported only a limited local reproduction.

Finally, AI-assisted code generated more figures and tables than the report required. I reviewed them against the research question and retained the smallest set that established correctness, estimator performance, Greek stability, and computational scalability. AI improved implementation speed, but the acceptance, rejection, interpretation, and selection of its outputs were my decisions.

Section 3: Reflections

Explain how you ensured (a) academic integrity, (b) accuracy/verification, and (c) **clear authorship** while using AI. Include one concrete example for each.

I addressed academic integrity by keeping AI separate from the project's evidential sources. I found and read the papers and market materials myself, and cited the original sources rather than treating ChatGPT as an authority. For example, I checked Lee et al.'s mathematical decomposition directly and distinguished its exact inclusion–exclusion structure from the logistic approximation used only for the unknown trivariate conditional probability. This prevented an AI explanation from being used as a substitute for the paper.

I ensured accuracy through a staged validation framework. Inputs and contract terms were frozen before the main experiments, and AI-assisted implementations had to pass analytical, published, or independently simulated benchmarks. I also used repeated randomisations, probability-bound and cash-flow checks, pre-specified tolerances, and CPU/GPU numerical-equivalence tests. A concrete example is that I reported GPU speed-up only after float64 CPU and GPU results for prices, Greeks, and conditional valuations agreed within the required tolerances. I measured warmed runtime separately from initial compilation and data-transfer costs, so the performance conclusion did not confuse speed with numerical correctness.

Clear authorship was maintained through my control over the research narrative and final wording. I decided the research question, selected the literature, extracted the relevant arguments, defined the tests, interpreted failures, chose the final evidence, and wrote the report. AI-generated code produced numerous diagnostic outputs, but their existence did not determine their importance. For example, I selected the combined CPU/GPU equivalence and scalability figure because it supported two claims central to the project, while leaving repetitive timing and diagnostic plots outside the main text. I also separated the real HSBC maturity-only note from the stylised continuously monitored research contract after checking the source data. These choices reflect my interpretation of what the evidence could legitimately establish, rather than an unedited AI-generated narrative.

Section 4: Additional information you may wish to include

All market data and academic sources were located and reviewed by me. All numerical results, tables, and figures were obtained by executing and checking the project notebooks. AI-generated code and outputs were treated as provisional until independently validated.

Declaration: I acknowledge the use of Chatgpt. I confirm that no content generated by AI has been presented as my own work (**please do not add your name or CID here and only the date**):

Date.....2026/08/11.....

Executive summary

This report investigates whether randomized quasi-Monte Carlo (RQMC) and Brownian-bridge conditioning improve the valuation and Greek estimation of a three-asset worst-of autocallable. Four estimators are compared: standard Monte Carlo (M0), direct RQMC (M1), Brownian-bridge-conditioned Monte Carlo (M2), and a conditioned RQMC hybrid (M3). The framework is validated against Black–Scholes results, an exact single-asset Brownian-bridge benchmark, and published autocallable cases. A Lee-style logistic approximation is reproduced but proves insufficiently robust outside its calibration setting, so the final conditioned methods use nested bridge estimates.

For the stylised market-informed contract, M3 achieves the highest pricing precision at large simulation budgets and gives a fair coupon of approximately 9.68%. However, no estimator dominates across all objectives. Standard Monte Carlo remains competitive at moderate accuracy requirements, while conditioning and RQMC are more effective at tighter tolerances. Greek performance is strongly trigger-dependent: M3 improves precision near the knock-in barrier, whereas direct simulation can be more stable near autocall triggers. GPU implementation further reduces high-budget computational cost. Overall, estimator choice should depend on precision requirements and trigger location rather than a universal ranking.

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1 Introduction

Autocallables combine early redemption, coupons, a worst-of basket and downside protection that changes discontinuously when barriers are crossed. Monte Carlo is flexible enough to represent these features, but its $O(N^{-1/2})$ convergence becomes expensive when binary triggers dominate the payoff; finite-difference Greeks can remain unstable even when price error appears acceptable (Glasserman, 2004; Broadie and Glasserman, 1996). This makes the product a useful test of whether smoother estimators deliver operational, rather than merely theoretical, gains.

The report asks:

For a three-asset worst-of autocallable, when do scrambled-Sobol sampling and Brownian-bridge conditioning reduce the time required to reach a target price accuracy, and do those gains extend to Delta, Vega and Gamma?

The empirical design separates three effects. First, M0–M3 identify the individual and joint contributions of RQMC and conditioning. Second, near-KI and near-autocall experiments distinguish the continuous barrier from the discrete redemption trigger. Third, matched CPU/GPU estimators separate statistical efficiency from hardware throughput. The contribution is therefore a reproducible decision framework across accuracy, stability and runtime, not a claim that either RQMC or conditional Monte Carlo is new.

Four subsidiary questions turn that framework into an auditable sequence. Does each engine reproduce analytical or published benchmarks? Does conditioning preserve value when a simulated crossing indicator is replaced by its conditional expectation? At matched wall-clock cost, where does the M0–M3 ranking change? Finally, are any price gains robust for bumped Greeks and large GPU batches? Each question has a frozen gate and a deliberately narrower claim than the headline comparison. This prevents a low standard error from being mistaken for low bias, or a fast kernel from being mistaken for a fast valuation workflow.

The evaluation unit is not an abstract variance ratio but a model output linked to an economic decision. Price uncertainty propagates directly into the fair coupon through $\delta C^* \approx -\delta V/A_C$, so a lower pricing RMSE is useful only after the extra computation needed to obtain it is counted. Greeks pose a different requirement: a desk needs a sensitivity that is stable across reasonable bump sizes, not merely one replication with a small standard error. The report therefore treats price accuracy, fair-coupon precision, bump-shape stability and wall-clock cost as related but non-interchangeable criteria. This distinction is especially important for a worst-of note, where the identity of the active asset can switch as the basket approaches an observation barrier.

The market case and research claim are deliberately separated. HSBC security 40447DKU1 supplies the SPX/NDX/RTY basket and market context, but its knock-in is observed only at maturity. All continuous-barrier experiments use a distinct stylised research contract. Hence the numerical values are model outputs, not dealer quotes or evidence of issuer mispricing.

Decision takeaway

The central result is a ranking reversal. M0 is the pragmatic choice at moderate accuracy; M3 becomes valuable when high precision is required. Conditioning helps most when the discontinuity it integrates out—the continuous knock-in—is economically active. It cannot by itself stabilise a separate discrete autocall trigger.

2 Literature and Research Gap

Scrambled digital nets retain unbiasedness and allow error estimation across independent randomisations, while their low discrepancy can improve effective convergence for sufficiently smooth, low-effective-dimensional integrands (Owen, 1997; L'Ecuyer, 2009; Joe and Kuo, 2008). Autocallable indicators violate that smoothness. This motivates combining RQMC with conditioning rather than assuming that a better point set alone will stabilise a discontinuous payoff.

For barriers, one-step survival changes the sampling distribution and attaches likelihood weights (Glasserman and Staum, 2001); Brownian-bridge probabilities instead integrate over crossings between simulated endpoints, removing the continuous-monitoring bias of a coarse grid (Broadie et al., 1997). For discrete autocall triggers, conditional analytic Monte Carlo and coordinate rotations can smooth a threshold after isolating an active normal coordinate (Fries and Joshi, 2011; Rakhmonov and Rakhmonov, 2019; Wang, 2016). These are different operations: continuous-KI conditioning does not automatically smooth coupon or autocall indicators.

The multivariate obstacle is the three-way exit probability. Let $A_i = \{M_i > b_i\}$ denote that bridge i crosses its barrier, conditional on terminal endpoint $\mathbf{X}(T) = \mathbf{x}$. The marginal exit is analytical,

$$g_i(\mathbf{x}) = P(A_i \mid \mathbf{X}(T) = \mathbf{x}) = \exp\left[-\frac{2b_i(b_i - x_i)}{\sigma_i^2 T}\right], \quad x_i \leq b_i, \quad (1)$$

and Lee et al. (2024) obtain pairwise co-exits $h_{ij} = P(A_i \cap A_j \mid \mathbf{X}(T) = \mathbf{x})$ through a two-dimensional wedge transformation and a modified-Bessel series. The only term without a simple closed form is $q_3 = P(A_1 \cap A_2 \cap A_3 \mid \mathbf{X}(T) = \mathbf{x})$.

Lee's key step is the exact factorisation

$$q_3 = h_{12}L_3(\mathbf{x}), \quad L_3 = P(A_3 \mid A_1 \cap A_2, \mathbf{X}(T) = \mathbf{x}). \quad (2)$$

Approximation enters only through the unknown conditional factor,

$$\hat{L}_3(\mathbf{x}) = \{1 + \exp[-(\alpha_0 + \boldsymbol{\alpha}^\top \mathbf{x})]\}^{-1}, \quad \hat{q}_3 = h_{12}\hat{L}_3. \quad (3)$$

This narrows the learning problem and preserves $0 \leq \hat{q}_3 \leq h_{12}$. Finally, inclusion–exclusion converts the exit terms into joint survival,

$$p_3 = 1 - \sum_{i=1}^3 g_i + \sum_{i < j} h_{ij} - q_3. \quad (4)$$

The construction is therefore semi-analytical rather than a direct regression of q_3 . Its local structure is attractive, but a match at one endpoint does not establish robustness across the state-dependent barriers encountered inside an autocallable.

For application to a live interval, the current log-price endpoint is translated to the bridge origin and each absolute knock-in barrier becomes a remaining distance from that endpoint. Interval length, the endpoint vector, volatilities, barriers and correlation together define an effective *signature*. An autocallable visits many such signatures as assets and remaining maturities change. Consequently, coefficients fitted inside one endpoint box cannot be assumed valid for every later interval. This mapping is the practical link between Lee's single-interval derivation and the decision to build a fresh nested reference whenever the production signature changes.

Autocallable studies offer analytical, PDE, Monte Carlo and Markov-chain approaches (Deng et al., 2011; Guillaume, 2015; Cui et al., 2024); simulation-Greek research also shows that discontinuities require specialised estimators (Gerstner et al., 2020). The remaining gap is an equal-cost comparison that treats three-asset bridge approximation error, price RMSE, Greek bump stability and end-to-end runtime as separate objects. The present study closes that narrower gap under a frozen correlated-GBM specification.

That cost normalisation is essential. A variance-reduction factor compares estimators at a fixed path count, but a conditioned path is more expensive because it requires bivariate special functions and a trivariate probability. The operational tests are therefore time to target RMSE and variance multiplied by total cost (Glynn and Whitt, 1992). GPU work is conceptually orthogonal: massively parallel simulation can move the runtime coordinate without changing the estimator or its bias (Abbas-Turki et al., 2014; Collange et al., 2015; Bilokon et al., 2022). Keeping statistical and hardware gains separate is what makes the final method choice usable rather than a collection of unrelated variance ratios.

3 Contract and Numerical Framework

3.1 Scope, payoff and market inputs

Table 1: Three contract layers keep market interpretation separate from numerical testing.

Layer	Identifier	KI monitoring	Role in the report
Market case	HSBC-40447DKU1	Maturity only	Product context and SPX/NDX/RTY inputs; no continuous-KI claim
Literature case	RC-L	Continuous	External validation against published pricing examples
Applied case	RC-A	Continuous	Stylised two-year worst-of research contract and main experiments

RC-A has quarterly observations, a frozen 8.75% annual coupon, a 100% autocall barrier, a 75% coupon and continuous knock-in barrier, coupon memory and a worst-of maturity loss. Let $R_{i,t} = S_{i,t}/K_i$ and $W_t = \min_i R_{i,t}$, where the contractual reference K_i remains fixed under spot bumps. Cash flows are audited as coupon value, early principal, surviving notional and terminal KI loss. Because coupon cash flows are linear in annual coupon C , the model fair coupon is

$$V(C) = V_0 + C A_C, \quad C^* = \frac{100 - V_0}{A_C}. \quad (5)$$

Under the risk-neutral measure, the three assets follow correlated GBM,

$$\frac{dS_i}{S_i} = (r - q_i)dt + \sigma_i dW_i, \quad dW_i dW_j = \rho_{ij} dt. \quad (6)$$

Exact lognormal transitions remove Euler bias. The 27 July 2026 snapshot uses 12-month ATM volatilities and dividend yields from the verified Bloomberg workbook, a transparent SOFR rate proxy, and 251 aligned daily observations for correlation. This hybrid of implied marginals and historical dependence is a controlled numerical specification, not full risk-neutral calibration.

3.2 Four estimators and the conditioning boundary

Table 2: The 2×2 estimator design isolates sampling and smoothing effects.

Outer sampling	Direct continuous-KI indicator	Bridge-conditioned KI weight
Pseudo-random	M0 — benchmark	M2 — conditioning effect
Scrambled Sobol	M1 — RQMC effect	M3 — hybrid estimator

M0/M1 simulate the knock-in on an explicit fine grid. M2/M3 simulate only contractual endpoints and replace the continuous-KI indicator by cumulative survival weights. For live interval k , the bridge probability $p_{3,k}$ from equation (4) gives $W_k = \prod_{j \leq k} p_{3,j}$. If L is the maturity downside loss and P_{base} is the payoff assuming par maturity principal, the conditioned payoff is

$$P_{\text{cond}} = P_{\text{base}} - L + LW_K. \quad (7)$$

Discrete coupon and autocall decisions remain indicators.

The project implementation of Lee’s logistic q_3 matches the reported local value, but fails the pre-specified out-of-sample RMSE gate. It is therefore not reused. Production M2/M3 construct a fresh nested scrambled-Sobol bridge bank for each interval signature, anchor it to the exact h_{ij} , and project raw q_3 estimates onto feasible probability bounds. The repair is recorded as a possible finite-inner-budget bias, not dismissed as floating-point cleaning.

Conditioning is a Rao–Blackwell step only when the interval survival probability is evaluated exactly: replacing a Bernoulli crossing event by its conditional expectation then preserves value while removing indicator noise. Here, g_i and h_{ij} are semi-analytical, whereas the trivariate term is numerical. Outer-path variance and inner-bank approximation error must therefore be audited separately. Increasing the outer budget cannot correct a biased q_3 , and probability projection guarantees feasibility rather than unbiasedness. Signature-specific banks reduce the extrapolation risk exposed by the logistic experiment, while paired Bernoulli-versus-weighted checks test whether the resulting survival weight preserves the direct valuation. This design is more expensive than reusing one fitted surrogate, but it makes the source of any disagreement observable and includes that additional work in the reported M3 runtime.

Central finite differences with common random numbers estimate Delta, Vega and Gamma. Spot bumps are 0.10%–1.00%, volatility bumps are 0.25–1.00 points, and absolute contractual references and barriers remain fixed. A separate active-coordinate diagnostic smooths only the next autocall principal; it tests the remaining discontinuity but is not a fifth full-product estimator.

The GPU implementation preserves the same float64 inputs, payoff conventions and batched common-random-number bumps. Transfer, path generation, payoff and reduction are included in timing. Exact Bessel evaluation for M2/M3 remains on the CPU, so conditioned valuation is explicitly a hybrid pipeline.

4 Validation and Experimental Design

Validation is staged so that a failure can be assigned to sampling, barrier monitoring, multivariate approximation or product logic rather than hidden by a plausible final price.

The principal accuracy–cost experiment evaluates $N \in \{2^{10}, 2^{12}, 2^{14}, 2^{16}\}$ using independent seeds

or Owen scrambles. Efficiency is reported as RMSE at equal wall-clock cost, variance multiplied by total cost, and time to a target RMSE (Glynn and Whitt, 1992). The RQMC-conditioning interaction ratio tests whether their joint gain exceeds the product of their separate gains. Across the completed study, 1,112 formal replications pass all 10 experiment gates and 25 tests.

The Greek design is intentionally local. The near-KI state is (0.765, 1.000, 1.000), placing asset 1 only 1.5 percentage points above the continuous barrier. The near-autocall state is (1.004, 1.001, 0.998), two days before the next observation, with asset 3 as the active worst name. M0 and M3 use 512 outer paths and four replications per state; M3 uses a fresh 128×12 inner bridge bank. Base and $\pm h$ valuations share a seed or scramble. The 0.10% spot bump is retained as a stress point but excluded from the strict bump-shape plateau because it is deliberately noise-dominated. Overlapping confidence intervals are recorded separately and cannot convert a failed shape test into a pass.

Timing follows the same discipline. GPU value and Greek gates use identical float64 inputs and include host-to-device transfer, path evolution, payoff evaluation and reduction. First invocation and warmed execution are separated, with at least three repeats and the median reported. The grid extends to $N = 2^{17}$, which makes fixed overhead, break-even and large-batch scaling visible rather than extrapolated.

Table 3: Validation ladder and frozen acceptance decisions.

Stage	Purpose and principal evidence	Decision
European option	Black–Scholes price and finite-difference Greeks validate MC/RQMC infrastructure	PASS
Single-asset bridge	Exact continuous-barrier benchmark separates sampling error from monitoring bias	PASS
Trivariate exit	Local target reproduced; OOS q_3 RMSE $0.003706 > 0.002500$	LIMITED
Direct product engine	Published RC-L price/coupon cases, cash-flow identity and probability mass	PASS
Conditioned product	M0–M3 agreement, nested-bank audit and paired Bernoulli/weight test	PASS
Greeks and GPU	CRN groups, bump grid, float64 equivalence and end-to-end timing	PASS

An independently randomised high-budget M3 run supplies the low-noise numerical reference 99.336888 (SE 0.001541; 95% CI [99.333868, 99.339908]). It shares M3’s endpoint and nested bridge construction, so it is not estimator-independent truth. Point-reference RMSE does not add the reference SE; removing that variance in quadrature changes the reported M3 RMSE from 0.011833 to 0.011732, without changing the ranking. M0 agreement, paired Bernoulli/weight comparisons and a high-budget nested audit reduce—but cannot eliminate—the circularity risk.

5 Results and Discussion

The preliminary gates explain which gains can be attributed to sampling and which require payoff smoothing. At $N = 65,536$, the European-call RQMC/MC relative-RMSE ratios are 0.127, 0.384, 0.144 and 1.127 for price, Delta, Vega and Gamma, respectively. RQMC therefore improves the

level of error for price and first-order sensitivities but does not automatically stabilise a second difference. In the exact single-asset bridge benchmark, the continuous price is 4.916410 and the weighted-RQMC maximum bias is 0.008423. By contrast, direct endpoint-monitoring bias falls only from 1.712233 at 12 steps to 0.315855 at 504 steps. Increasing N reduces sampling error but cannot remove this separate monitoring bias.

5.1 Pricing validity and fair coupon

The direct engine stays within every external tolerance: the maximum RC-L Table 7 price error is 0.761446/100, the maximum fair-coupon error is 0.010419, and the maximum Table 8 price error is 0.299435/100. For the frozen principal configuration, the independently randomised high-budget M3 value is 99.336888/100. Solving the same conditional model for par gives $C^* = 9.684969\%$. The first number is the price at the contractual coupon; the second is a root of equation (5). They are not two competing estimates of the same quantity.

Table 4: Pricing scorecard: external validity, internal agreement and model output.

Evidence	Observed result	Benchmark / reading	Status
RC-L published prices	Max error 0.761446	Tolerance 1.25 per 100	PASS
RC-L published coupon	Max error 0.010419	Tolerance 0.0200 in rate units	PASS
Step-down published prices	Max error 0.299435	Tolerance 1.25 per 100	PASS
M0–M3 agreement	Max difference 0.356807	Tolerance 1.25 per 100	PASS
Principal M3 value	99.336888 (SE 0.001541)	Low-noise numerical reference	MODEL
Principal fair coupon	9.684969%	Conditional GBM root $V(C^*) = 100$	MODEL

The high-budget M3 value is an independently randomised numerical reference, not an exact solution. Its 95% confidence interval is [99.333868, 99.339908], and the evaluation runs do not reuse its scrambles. However, independence removes sample reuse rather than estimator-specific approximation bias. Reported RMSE is measured against the reference mean and does not add the reference standard error mechanically to every observation. Agreement with high-budget M0 and the paired Bernoulli construction is therefore used as a directional cross-check; the remaining common-model and nested-probability risk is retained in the limitations.

The trivariate gate provides a second boundary on interpretation. The local Lee-style value 0.165739 is close to the reported 0.1631 and the exact bivariate terms are numerically stable, yet the seven-endpoint out-of-sample RMSE is 0.003706, with maximum error 0.006158. Changing the training grid from 25 to 100 and 1,000 steps moves the selected logistic estimate from 0.165739 to 0.175380 and 0.181632. The failure is therefore approximation and specification risk rather than a probability-bound error, which is why the production estimator retains the exact decomposition but replaces the fitted conditional factor.

The validation comparisons establish implementation credibility rather than a market calibration. M0–M3 agree across both research contracts: the maximum difference from M0 is 0.356807/100,

below the 1.25 tolerance. The main price and coupon are therefore reported once, under the final accuracy–cost configuration, while stage-specific intermediate levels are used only to compute the validation gates below. This avoids promoting diagnostic configurations into additional economic valuations.

The nested q_3 audit has maximum RMSE 0.000532 and maximum error 0.002604, with no final probability-bound violations. RC-A projection is frequent (24.22%–24.55%) but usually tiny: mean absolute repair is about 1.7×10^{-4} , the largest 99th percentile is 0.008775 and the maximum single repair is 0.058331. The high-budget audit suggests a small aggregate effect, but the nonlinear repair remains a source of approximation bias.

5.2 Greeks depend on the active trigger

Near the continuous 75% knock-in, M3 reduces replication SD for active Delta, Vega and Gamma by factors of 1.31, 1.87 and 2.05. Two days before an autocall observation, however, KI conditioning leaves the discrete trigger unchanged: M0 is more stable for active Delta and Gamma, while M3 still improves Vega.

Table 5: Active-asset precision at the middle bump; ratios above one favour M3.

Trigger region	Greek	M0/M3 SD	Winner	Reading
Near KI	Delta	1.31	M3	Bridge event is active
Near KI	Vega	1.87	M3	Strongest first-order gain
Near KI	Gamma	2.05	M3	Lower SD, but no reliable plateau
Near autocall	Delta	0.34	M0	Discrete trigger dominates
Near autocall	Vega	1.79	M3	Gain survives for Vega
Near autocall	Gamma	0.37	M0	M3 amplifies bump noise

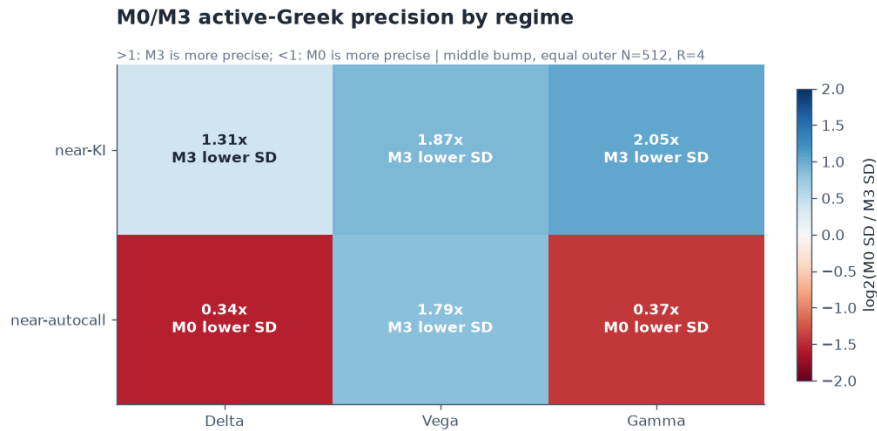


Figure 1: M0/M3 Greek-precision ratios show a trigger-dependent ranking reversal.

The strict full-product Gamma bump-shape plateau rate is only 12.5%. Confidence intervals may overlap across bumps, but that weaker condition does not demonstrate a flat numerical region.

Gamma is therefore retained as an instability diagnostic, not presented as a dependable hedge estimate. This negative result matters: lower estimator variance at one bump does not remove finite-difference bias or switching noise.

This reversal follows from the algebra of finite differences. Delta divides a price difference by h , while Gamma divides a second difference by h^2 ; any trigger-switching noise is therefore amplified as the bump shrinks. Common random numbers cancel smooth pathwise variation, but they do not prevent a path from redeeming in one bumped valuation and continuing in another. Near knock-in, M3 integrates out precisely the crossing event that causes this mismatch. Near autocall, the dominant discontinuity is instead the loss of future coupons and continuation value at a discrete date, which the KI weight leaves untouched. For that reason, replication SD at one chosen bump is insufficient evidence of a hedge-quality Greek: a credible estimate should also exhibit a stable region across adjacent bumps. The failed Gamma plateau is thus a substantive finding about estimator scope rather than a failed presentation metric.

Local active-coordinate smoothing identifies the missing mechanism. Across all 15 spot/horizon cells, the next-observation component agrees with raw RQMC and reaches a maximum variance-reduction factor of $29,858\times$. In the focused experiment, SD falls by 97.0, 61.2 and 126.1 times for price, active Delta and active Gamma. These are diagnostic gains only: coupons, later observations and terminal cash flows are outside this local estimator.

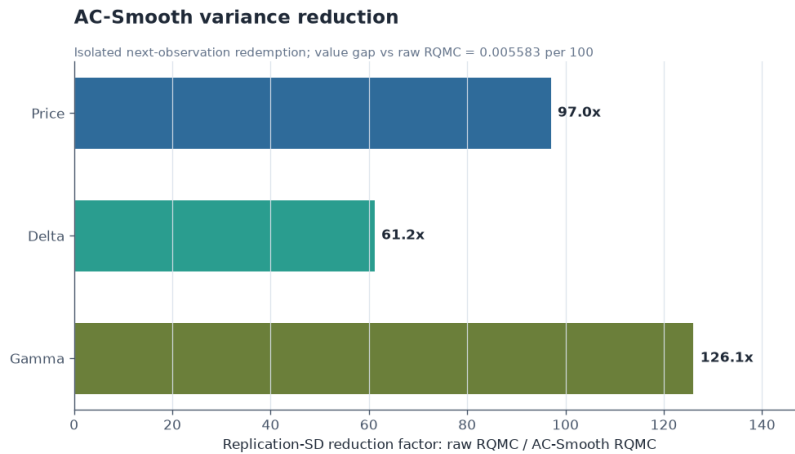


Figure 2: Local trigger smoothing sharply reduces dispersion for the isolated next-observation redemption component.

5.3 Accuracy, cost and implementation choice

At $N = 65,536$, M3 attains RMSE 0.011833 and lies on the accuracy–cost Pareto frontier. M0 nevertheless reaches a moderate target RMSE of 0.05 faster. The RQMC–conditioning interaction ratio ranges from 0.950 to 2.746 and exceeds one at three of four budgets: the methods are usually complementary, not universally so. This result is stronger than a variance-reduction-factor claim because fresh nested-bank construction is included in M3 cost.

This creates a practical two-threshold policy. Below the moderate error target, M0’s low setup cost dominates; at tighter tolerances, M3’s smaller dispersion repays the bridge overhead. The observed frontier is empirical, not an asymptotic guarantee: it depends on the tested budgets, hardware and frozen contract. Nevertheless, it is more actionable than reporting that one method has lower variance at the same N , because it answers which implementation reaches a business accuracy requirement first.

GPU correctness is established before speed is discussed. The maximum absolute CPU/GPU

difference is 1.42×10^{-14} for direct value and 4.45×10^{-10} across batched Delta, Vega and Gamma; all 9 validation gates and 19 tests pass. Warmed end-to-end break-even occurs near $N = 8,192$, and the direct pipeline is $9.28\times$ faster at $N = 131,072$. Because the exact Bessel stage remains on CPU, this speed-up does not transfer automatically to the full conditioned pipeline.

Table 6: Decision matrix: estimator choice follows the task, not a single ranking.

Decision context	Preferred route	Evidence and boundary
Moderate price accuracy	M0	Faster at RMSE 0.05; transparent and cheap to operate
High-precision price	M3	RMSE 0.011833 at $N = 65,536$; Pareto-efficient after nested cost
Near continuous KI	M3	Lower Delta/Vega/Gamma SD because the active event is conditioned
Near discrete autocall	M0 + local diagnostic	M0 wins Delta/Gamma; local smoothing shows where a full extension should act
Large direct workloads	GPU pipeline	Numerical equivalence; $9.28\times$ speed-up at $N = 131,072$

RC-A is a market-informed research contract under constant-volatility GBM. ATM volatilities omit skew and term structure; historical correlation is sampling- and regime-sensitive; issuer credit, funding, liquidity and dealer margin are excluded. Fair coupons and estimator rankings are conditional on the frozen model, not executable market recommendations.

6 Conclusion and Limitations

The validation chain supports the pricing engine while rejecting an overly broad claim for the simple logistic trivariate surrogate. Fresh nested bridge conditioning gives M0–M3 price agreement and lets M3 access a high-precision region that M0 does not reach as efficiently at the tested budgets. It is not a universal replacement: M0 remains the faster moderate-accuracy choice, and RQMC–conditioning interaction is budget-dependent.

The Greek evidence sharpens the conclusion. M3 is effective near the event it smooths—continuous knock-in—but does not regularise the discrete autocall trigger. The 12.5% Gamma plateau rate prevents interpreting full-product Gamma as a reliable hedge quantity. Local trigger conditioning shows substantial potential, yet it must be extended across every coupon and redemption date before it can become a complete valuation estimator.

GPU acceleration is a separate engineering gain: float64 equivalence is preserved and large direct workloads become materially faster, while M2/M3 retain a CPU special-function boundary. The resulting workflow is precision-specific and trigger-aware: direct simulation for moderate-accuracy tasks, conditioned RQMC for demanding price accuracy, and a separate treatment for near-autocall Greeks.

A Reproducibility Protocol

This appendix records the implementation choices needed to reconstruct the reported comparisons. It is intentionally more detailed than the main text: the numerical conclusions depend on matching both the estimator and its cost boundary.

Table 7: Frozen implementation and timing protocol.

Item	Implementation detail
Software	Python 3.13.5; NumPy/SciPy/pandas on CPU and CuPy/CUDA on GPU. Exact package, driver and device identifiers were not retained, so absolute timing portability remains limited.
RQMC	Independent Owen-scrambled Sobol replications; 36 time-major coordinates for 12 endpoints and three assets; uncertainty estimated across scrambles.
Budgets	Outer $N \in \{2^{10}, 2^{12}, 2^{14}, 2^{16}\}$; separate high-budget reference; GPU scaling through 2^{17} .
CRN policy	Base and $\pm h$ repricing share one seed or Sobol scramble within each replication; contractual references and absolute barriers remain fixed.
Nested bridge	Fresh bank per interval signature; production cost includes bank construction, Bessel evaluation, q_3 estimation, projection, weighting and reduction. Audit bank $4,096 \times 64$ uses independent seeds.
GPU boundary	Transfer, path evolution, payoff and reduction included; first call separated from warmed execution; at least three repeats and the median reported. M2/M3 special functions remain on CPU.
Reference	High-budget M3 reference and evaluated runs use independent randomisations. Point-reference RMSE excludes reference SE and is interpreted subject to the same-estimator caveat.
Audit trail	Executed notebooks retain frozen configurations and replication tables; 1,112 formal replications, 10/10 gates and 25/25 tests pass.

A.1 Experimental budgets and uncertainty units

Table 8: Frozen experiment grid used in the validation chain.

Experiment	Budget and replications	Principal output
European calls	$N = 2^{10}-2^{16}$, 16 seeds or scrambles	Price and Greek relative RMSE against Black–Scholes
Exact one-asset bridge	$M = 12-504$, $N = 2^{10}-2^{16}$	Sampling error separated from monitoring bias
Trivariate exit gate	Appendix-C point, seven OOS endpoints and eight published cases	Logistic approximation error and fallback decision
Direct product	RC-L: $N = 2^{12}$, $R = 3$; RC-A: $N = 2^{12}$, $R = 4$	External prices, cash-flow identities and coupon roots
Conditioned product	$N = 2,048$; M0/M1 $R = 4$, M2/M3 $R = 6$	M0–M3 price agreement and nested-bank audit
Greek diagnostic	$N = 512$, $R = 4$; M3 bank 128×12	Two trigger regions and complete bump grids
Accuracy–cost study	$N = 2^{10}, 2^{12}, 2^{14}, 2^{16}$	RMSE, total time, interaction and Pareto frontier
GPU scaling	Matched float64 inputs through $N = 2^{17}$	Correctness, first-call/warmed time and break-even

For a low-noise reference V_{ref} , reported sampling accuracy is

$$\text{RMSE}(\hat{V}) = \left\{ R^{-1} \sum_{r=1}^R (\hat{V}_r - V_{\text{ref}})^2 \right\}^{1/2}.$$

Independent seeds are used for MC and independent Owen scrambles for RQMC. Within a Greek replication, however, base and bumped valuations deliberately share the same randomisation. This common-random-number coupling is part of the estimator and is kept identical across methods. Work-normalised error is reported as $\text{RMSE}^2 \times \text{total runtime}$. For conditioned methods, total runtime includes construction of the nested bridge bank; for GPU methods it includes transfer, evolution, payoff evaluation and reduction. These definitions prevent a kernel-only or path-count-only comparison from being presented as an operational efficiency result.

B Validation Evidence

B.1 Analytical and single-asset benchmarks

Table 9: European engine validation at $N = 65,536$: mean relative RMSE across SPX, NDX and RTY.

Quantity	MC	RQMC	Ratio	Reading
Price	0.006638	0.000846	0.127	Material RQMC gain
Delta	0.004179	0.001606	0.384	Material RQMC gain
Vega	0.008780	0.001261	0.144	Material RQMC gain
Gamma	0.023707	0.026726	1.127	No RQMC advantage

Table 10: Exact continuous-barrier benchmark and monitoring-bias audit.

Diagnostic	Observed result	Interpretation
Continuous benchmark	Analytic price 4.916410	Target for bridge-corrected estimators
Weighted bridge RQMC	Maximum absolute bias 0.008423	Passes 0.025 frozen tolerance
Bernoulli vs weight	Maximum paired z -score 2.023	Conditional expectation preserves value
Endpoint monitoring	Bias 1.712233 at $M = 12$; 0.315855 at $M = 504$	Dense grids reduce but do not eliminate monitoring bias
Delta and Vega	Maximum errors 0.008690 and 0.003061	First-order sensitivity gates pass
Gamma	Cross-bump range 0.006852; max error 0.010150	Full disclosure retained; no strict plateau claim

These two benchmarks are deliberately complementary. The European test checks discounting, marginal GBM simulation, scrambling and finite differences in a smooth setting. The bridge test then introduces the exact discontinuity that M2/M3 are intended to integrate out. A better Sobol point set cannot correct a wrong monitoring convention, while a bridge correction does not guarantee a stable Gamma. Keeping those claims separate is central to the later autocallable interpretation.

B.2 Trivariate exit-probability gate

Table 11: Independent implementation of the Lee-style trivariate decomposition.

Criterion	Observed result	Decision
Local target	0.165739 vs 0.1631; error 0.002639	Passes the 0.005 local tolerance
Bivariate series	$h_{12} = 0.237504$, $h_{13} = 0.256498$, $h_{23} = 0.279012$	Stable from 12 to 40 Bessel terms
Published cases	RMS relative error 0.0071 vs 0.0552 for discretised direct MC	Supports exact marginal/pairwise implementation
Training uncertainty	Independent-seed SD 0.001667	Material and distinct from pricing SE
OOS accuracy	RMSE 0.003706; max error 0.006158	Fails the 0.002500 production gate
Probability bounds	Zero violations in 24,059 eligible audit endpoints	Valid probabilities do not imply accurate approximation
Overall	Exact factorisation retained; fitted L_3 rejected	LIMITED

The local pass and OOS failure are not contradictory. Equations (2)–(4) are exact probability identities, whereas the logistic coefficients approximate a state-dependent conditional factor. The autocallable visits endpoint and remaining-barrier signatures that differ from the training box. Production M2/M3 therefore preserve the analytical g_i , h_{ij} and inclusion–exclusion structure but generate a fresh nested estimate of q_3 for the signature being valued.

B.3 Contract engine and external price checks

Before stochastic pricing, ten deterministic paths cover early and later autocalls, missed and recovered memory coupons, no-knock-in maturity, continuous knock-in recovery, terminal loss, barrier equality and asymmetric worst-of outcomes. All ten intended outcomes are recovered; the cash-flow component identity has zero maximum error. A bracketed solver also recovers a deliberately chosen 7.5% coupon with error 2.78×10^{-17} , establishing the coupon interface independently of any market interpretation.

Table 12: Direct-product validation against published cases and internal identities.

Criterion	Observed result	Status / interpretation
Published RC-L values	Maximum error 0.761446 per 100	PASS
Published RC-L coupons	Maximum rate error 0.010419	PASS
Published step-down values	Maximum error 0.299435 per 100	PASS
Probability mass	Maximum absolute error 0	PASS
Component identity	Maximum absolute error 1.421×10^{-14}	PASS
Coupon roots	Linear residual 0; bracketed residual 3.881×10^{-11}	PASS

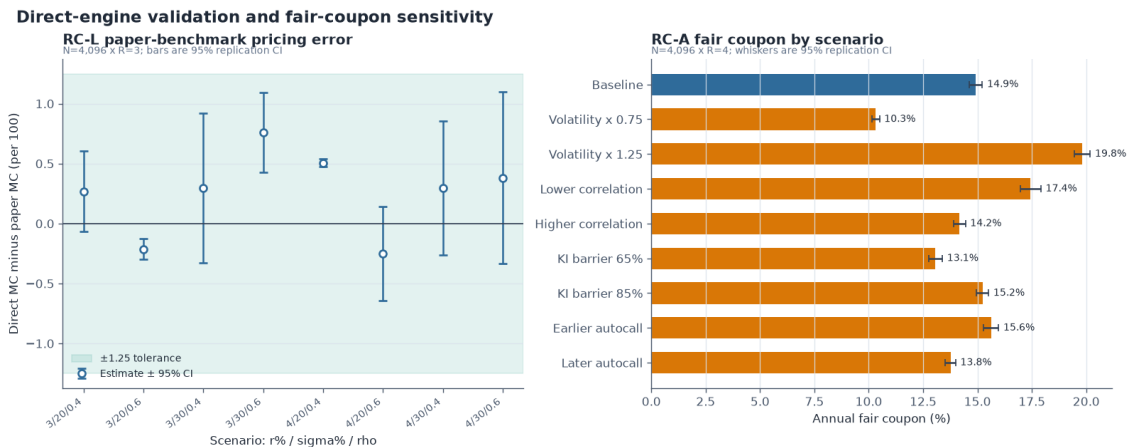


Figure 3: Published-case pricing errors and fair-coupon scenario sensitivity for the direct-engine validation configuration.

The sensitivity panel above belongs to the direct-engine validation configuration and should not be read as a second estimate of the final high-budget M3 fair coupon. Its role is to check economically sensible signs and the stability of the linear coupon root under one-factor changes.

C Conditioned-Estimator and Computational Audits

C.1 M0–M3 value agreement

Table 13: Stage-specific M0–M3 value agreement; values and SEs are per 100.

Contract	Method	Value	SE	Difference from M0
RC-L	M0	101.1512	0.1494	0.0000
RC-L	M1	101.2306	0.1551	0.0794
RC-L	M2	100.7943	0.0636	-0.3568
RC-L	M3	101.1592	0.0944	0.0081
RC-A	M0	93.7016	0.2589	0.0000
RC-A	M1	93.9698	0.0784	0.2682
RC-A	M2	93.9361	0.1890	0.2345
RC-A	M3	93.8895	0.0347	0.1879

These are equal-budget validation runs, not the high-budget principal value reported in the main text. Their purpose is to test whether replacing the direct bridge indicator by its conditional expectation changes the target. The maximum difference from M0 is 0.356807, below the frozen tolerance of 1.25 per 100; paired Bernoulli/weight comparisons also pass.

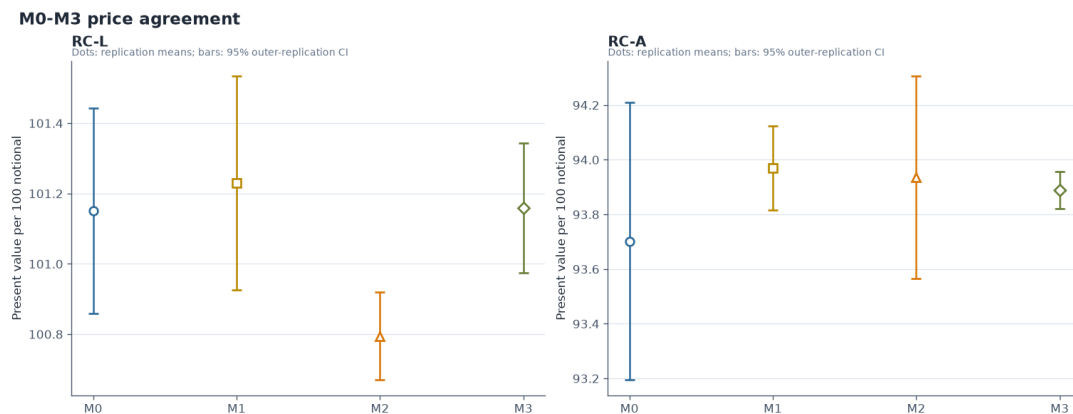


Figure 4: M0–M3 value agreement with 95% outer-replication intervals in the validation configuration.

C.2 Nested probability, reference and GPU audits

Table 14: Approximation, reference and implementation audit.

Audit item	Observed result	Interpretation
Fresh nested bank	RMSE 0.000532; max error 0.002604	Inner approximation tracked separately from outer sampling
RC-A q_3 projection	24.22–24.55% of raw values	Feasibility repair is common, not an exceptional event
Repair magnitude	Mean $\approx 1.7 \times 10^{-4}$; max 99th percentile 0.008775; maximum 0.058331	Usually small but nonlinear; potential finite-inner-budget bias
Final probability bounds	Zero violations and zero underflows	Repaired probabilities are feasible
Numerical reference	99.336888, SE 0.001541; 95% CI [99.333868, 99.339908]	Low-noise M3 reference, not estimator-independent truth
Reference-SE adjustment	M3 RMSE 0.011833 becomes 0.011732 after quadrature removal	Ranking is insensitive to reference uncertainty
CPU/GPU direct value	Maximum difference 1.42×10^{-14}	Float64 equivalence
CPU/GPU Greeks	Maximum difference 4.45×10^{-10}	Batched CRN implementation equivalent
GPU scale	Break-even near 8,192; $9.28\times$ at 131,072	Applies to the direct pipeline; Bessel stage remains on CPU

Projection has to be interpreted carefully. A high projection rate does not imply a large pricing distortion because most corrections are small; equally, a small mean repair does not prove unbiasedness because projection is nonlinear. The high-budget nested audit and M0 agreement constrain the likely aggregate effect, but the report retains probability repair as a modelled numerical risk. The same caution applies to the M3 reference: independent randomisation removes sample reuse, not shared-estimator bias.

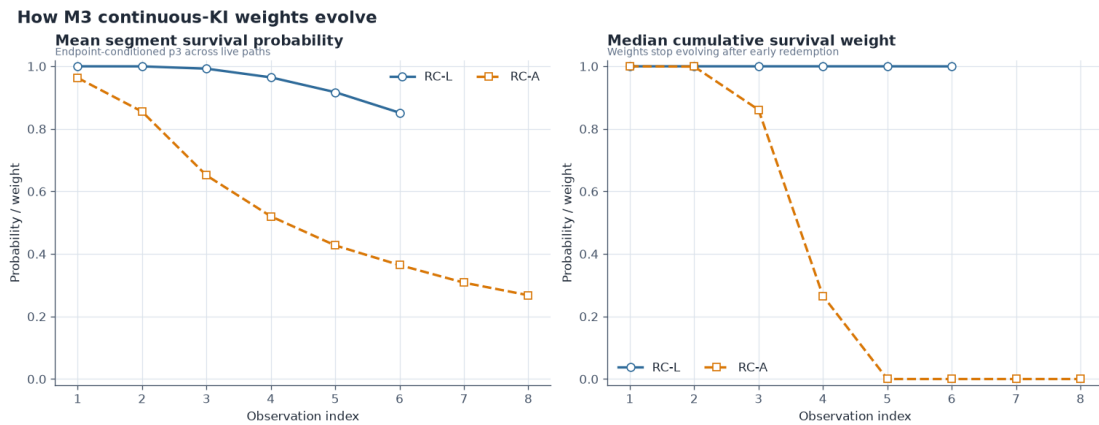


Figure 5: Segment and cumulative continuous-KI survival weights across live intervals.

D Supplementary Greek Diagnostics

The complete bump curves are retained to make the Gamma limitation auditable. The 0.10% spot point is shown but excluded from the strict plateau decision; uncertainty consistency is recorded separately from bump-shape stability.

Table 15: Active-asset replication SD at the middle bump.

Region	Greek	M0 SD	M3 SD	M0/M3	Lower SD
Near KI	Delta	0.1302	0.0993	1.31	M3
Near KI	Vega	0.0885	0.0473	1.87	M3
Near KI	Gamma	0.2146	0.1045	2.05	M3
Near autocall	Delta	0.1211	0.3540	0.34	M0
Near autocall	Vega	0.0526	0.0294	1.79	M3
Near autocall	Gamma	0.5798	1.5508	0.37	M0

The table reports dispersion, not economic hedge levels. In particular, the near-KI Gamma has lower M3 SD at the middle bump but still fails the strict cross-bump stability requirement in most configurations. Across the full product, only 12.5% of Gamma series form the pre-specified plateau. This is why the main conclusion treats Gamma as evidence about estimator instability rather than as a reliable position-level hedge.

Table 16: Boundary of the local discrete-trigger smoothing diagnostic.

Evidence	Observed result	Permitted interpretation
Value agreement	15/15 spot/horizon cells agree with raw RQMC	Active-coordinate integration preserves the isolated component
Maximum VRF	29,858×	Discrete-trigger smoothing can be extremely effective locally
Focused price SD	97.0× reduction	Applies to next-observation principal only
Focused Delta SD	61.2× reduction	Supports the proposed smoothing mechanism
Focused Gamma SD	126.1× reduction	Does not validate full-product Gamma
Missing cash flows	Coupons, later observations and terminal payoff excluded	Not a fifth full-product estimator

The diagnostic aligns the active trigger with one normal coordinate and integrates over that coordinate conditionally. Extending it to the full note would require a consistent treatment at every observation date, including coupon memory and continuation value. Until that recursion is implemented, the large local variance reductions diagnose where M0/M3 remain rough; they do not replace the product valuation or the reported near-autocall comparison.

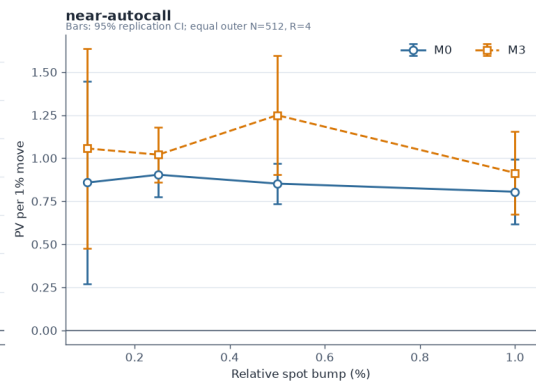
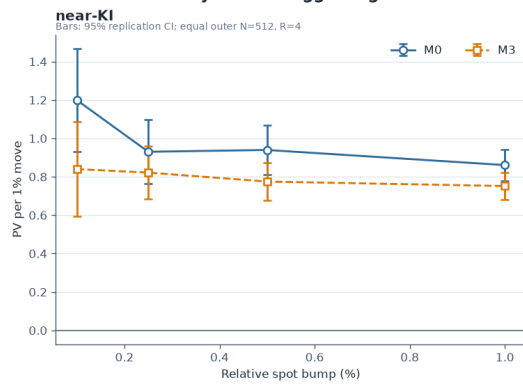
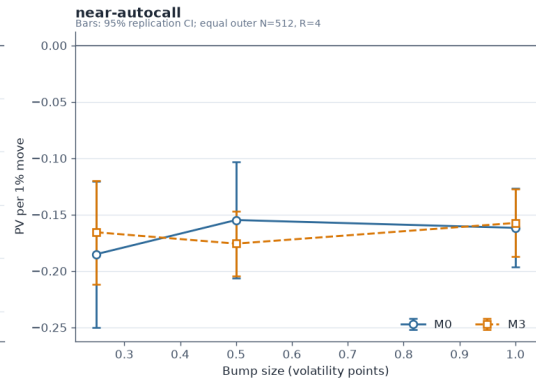
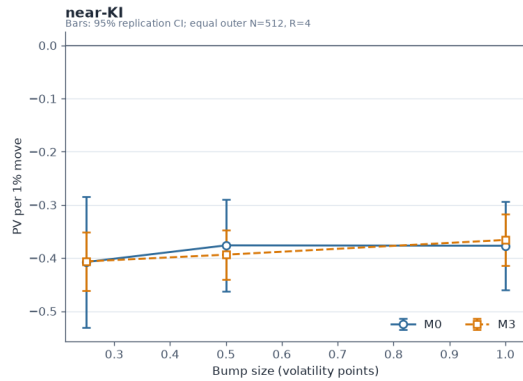
Active Delta stability across trigger regimes**Active Vega stability across trigger regimes**

Figure 6: Active-asset Delta (top) and Vega (bottom) across near-KI and near-autocall regimes.

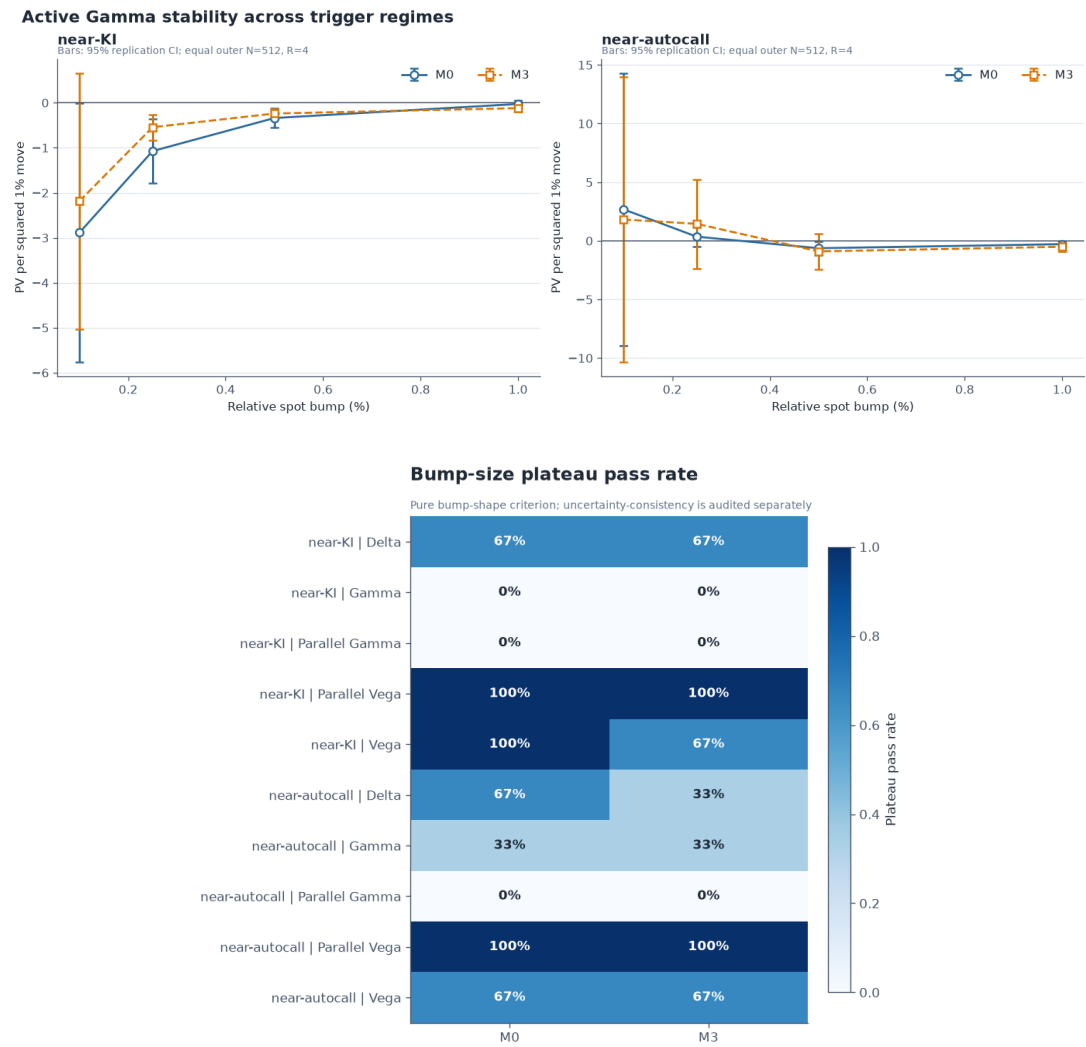


Figure 7: Active-asset Gamma curves (top) and strict bump-shape plateau audit (bottom).

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